College of Vocational Studies
University of Delhi
Cloud Computing
(Practical File)

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Practical 1: Cloud Platforms

a) Create free-tier accounts on AWS, Azure, and GCP.
b) Explore dashboards and identify key services (compute, storage, networking).
c) Understand pricing calculators on each platform.

<<INCLUDE STEPS, SCREENSHOTS, COMPARISON>>

Practical 2: Create a virtual machine on Microsoft Azure

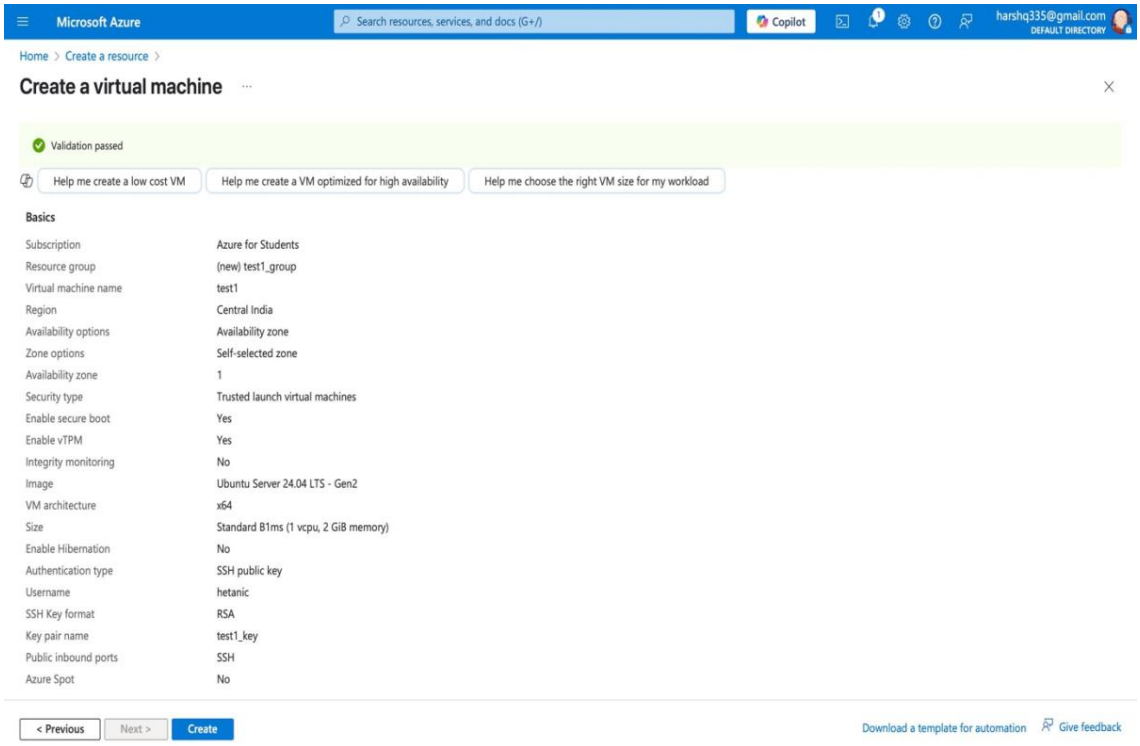
Step 1: Logging to Microsoft Azure portal using Username and Password.

Step 2: Create virtual machine

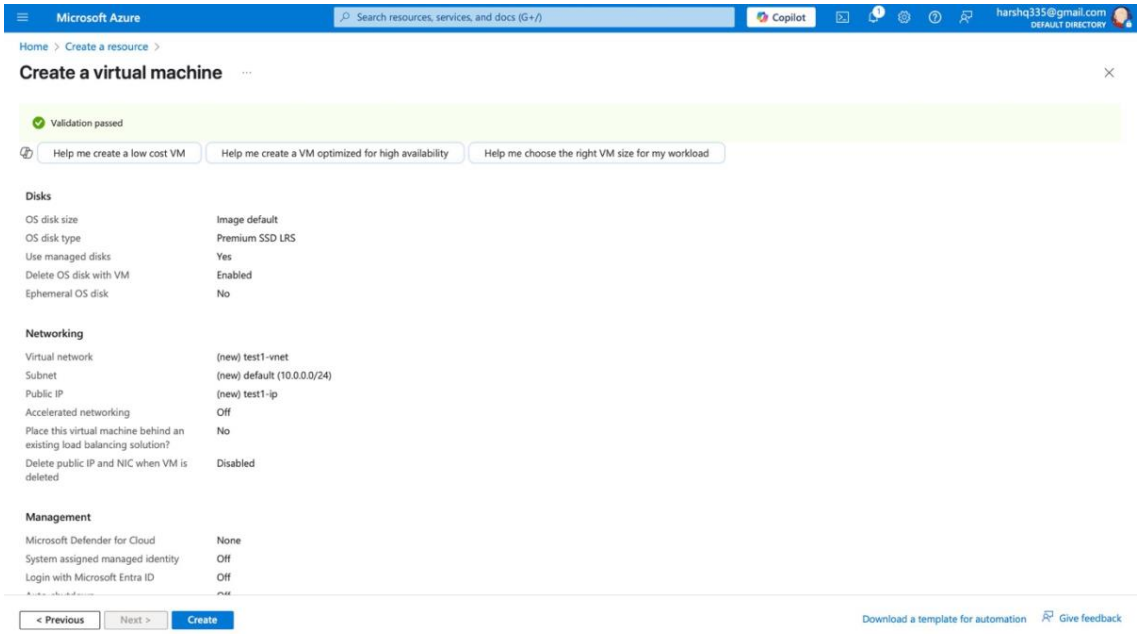
Enter virtual machines in the search. Under Services, select Virtual machines.

In the Virtual machines page, select Create and then Azure virtual machine. The Create a virtual machine page opens.

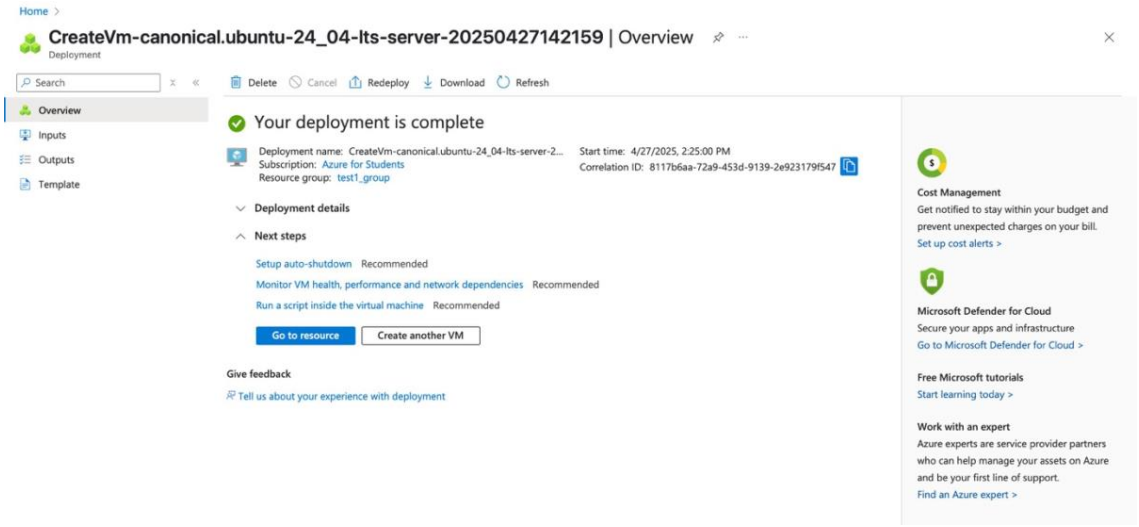
Under Instance details, enter "test1" for the Virtual machine name and choose "Ubuntu Server 24.04 LTS – Gen2" for the Image. Leave the other defaults.



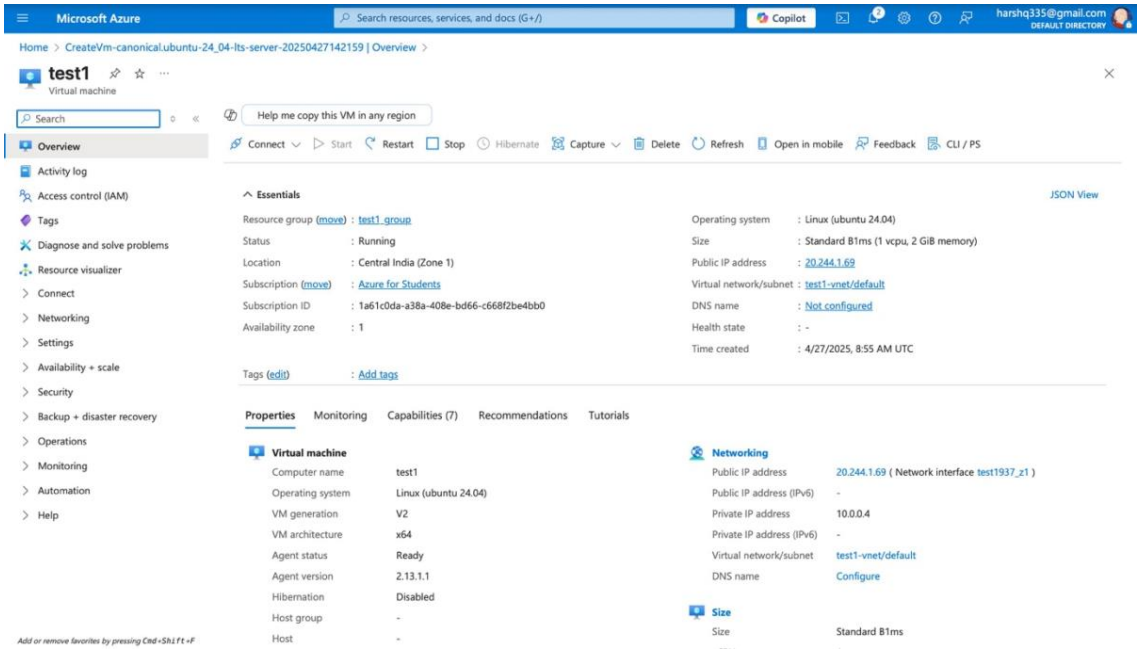
Select the OS disk type (Standard HDD/Standard SSD/Premium SSD). Add data disks if needed.



Click on Create to deploy the virtual machine.



Click on "Go to resource" to view the deployed virtual machine.



Practical 3: Launch Your First Amazon EC2 Instance

Step 1: Access the EC2 Console

Sign in to the AWS Management Console. You can search for "EC2" in the search bar or select "EC2" from the services menu.

Step 2: Launch an Instance

Click the "Launch Instance" button.

Use a pre-configured Amazon Machine Image (AMI) (e.g., Amazon Linux 2).

Select an instance type that meets your needs. For a beginner, t2.micro is a low-cost, general-purpose option.

Configure the instance like Network, Subnet, etc.

Configure security groups to allow SSH access.

Click on "Review and Launch" and then click on "Launch".

Create a new key pair and download. Click on "Launch Instances".

Step 3: Connect to the instance using SSH.

Select the machine instance and click on "Connect".

Click on "RDP Client" and click on "Download remote desktop file". Save the prompted file on local hard drive and click on the downloaded file. click on "Connect".

Enter Key-Pair created before and click on Ok.

Practical 4: Cloud security

Implement IAM roles and policies for a cloud platform

Roles are collections of permissions that define what actions a user, group, or service account can perform on cloud resources.

Policies are JSON documents that describe which actions are allowed or denied for a specific identity and resource.

An IAM role is an IAM identity that you can create in your account that has specific permissions. An IAM role is similar to an IAM user, in that it is an AWS identity with permission policies that determine what the identity can and cannot do in AWS.

You can use roles to delegate access to users, applications, or services that don't normally have access to your AWS resources. For example, you might want to grant users in your AWS account access to resources they don't usually have, or grant users in one AWS account access to resources in another account.

Roles can be assumed by the following:

- An IAM user in the same AWS account or another AWS account
- IAM roles in the same account
- Service principals
- An external user authenticated by an external identity provider (IdP) service that is compatible with SAML 2.0 or OpenID Connect

Defining Least Privilege

Grant each identity only the necessary permissions to perform their tasks.
For Example, A developer should have access to their development environment and its resources, but not to the production environment or sensitive data.